

Chapter 111 — Underactive Detrusor — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

- **Detrusor underactivity (DUA)** = urodynamic dx; **underactive bladder (UAB)** = symptoms
- **DUA on urodynamics (UDS)**: men <50 → 9–28%; men >70 → 48%; women → 12–45%
- **Bladder contractility index (BCI)** = $P_{det}@Q_{max} + 5Q_{max}$: >150 strong, 100–150 normal, <100 weak
- **Only accepted diagnostic test**: invasive UDS; **no consensus criteria**
- **Preferred drainage**: intermittent self-catheterisation (ISC); long term, **suprapubic catheter (SPC)** > **urethral catheter**
- **No effective drug**: parasympathomimetics avoided — **risk of cardiac arrest**
- **Sacral neuromodulation (SNM)** → DUA with sphincter / pelvic floor disorder
- **Outlet surgery, low PVR, symptoms only** → unlikely to help
- **Catheter-dependent man, young, fit** → outlet surgery reasonable
- **Preserved bladder sensation** = best predictor of treatment response

TERMINOLOGY, DEFINITIONS, SYMPTOMS

- **International Continence Society (ICS) 2002 DUA**: contraction of reduced strength and/or duration → prolonged or incomplete emptying; **no numeric thresholds**
- **No contraction** → **acocontractile detrusor**
- **ICS UAB (2018)**: slow stream, hesitancy, straining ± incomplete emptying ± storage symptoms
- **No pathognomonic symptom** (vs overactive bladder [OAB] = urgency)
- **Significant postvoid residual (PVR)** (one proposal): >40% of functional capacity (voided + PVR)
- **Sensation absent** → infrequent voider, nocturnal incontinence; **preserved** → frequency, nocturia

EPIDEMIOLOGY

- **Population prevalence unknown** — no non-invasive marker
- **Female bladder outlet obstruction (BOO)**: only 2.7–8% of UDS referrals → low flow + high PVR in woman ≈ DUA
- **Men >65–70**: DUA 40–48%; institutionalised elderly ≈ 40%
- **Detrusor hyperactivity with impaired contractility (DHIC)**: DO + DUA; frail/institutionalised elderly
- **Coexistence**: ~half of men with DUA also DO/BOO; ~¾ of women also DO/stress urinary incontinence (SUI) → **UDS before LUTS surgery in elderly**
- **Untreated DUA ($Q_{max} < 15$, $P_{det}@Q_{max} < 40$)**: minimal progression at 10 yr

ETIOPATHOGENESIS

- **Confirmed causes**: neurological disease/injury, **diabetes**
- **Probable only**: BOO, ageing, atherosclerosis/pelvic ischaemia
- **Mechanisms**: **myogenic** (myocyte/extracellular matrix) vs **neurogenic** (afferent, efferent, central)
- **Elbadawi degeneration pattern**: disrupted myocytes + axonal degeneration
- **Smith hypothesis**: stiff bladder → wall stress collapses early in void → reflex stops → PVR
- **BOO animal model**: compensation → decompensation (fibrosis) — irreversible if not relieved first
- **Mechanism**: Laplace → ischaemia-reperfusion → oxidative injury → denervation
- **Most men in acute retention**: detrusor preserved
- **High-pressure chronic retention**: preserved contractility, poor compliance, upper-tract change
- **Diabetic cystopathy**: early storage symptoms → late poor sensation + voiding failure
- **Acute stroke**: ~½ retain ("cerebral shock"); chronic = DO
- **Parkinson disease**: DUA <20%, often from **anticholinergic drugs**

- **Multisystem atrophy**: DUA 52–95% → retention
- **Multiple sclerosis**: DUA ~20% (lumbosacral plaques)
- **Radical hysterectomy**: LUTD ~72%; rectal surgery now <5%; many recover by 1 yr
- **Reversible**: Guillain-Barré, herpes zoster; **permanent**: AIDS, tabes dorsalis

DIAGNOSIS

- **$P_{det}@Q_{max} + Q_{max}$** underestimates strength: bladder outlet relation (BOR) → $P_{det}@Q_{max}$ is lowest-pressure point
- **Projected isovolumetric pressure (PIP, Schäfer)** = $P_{det}@Q_{max} + K \cdot Q_{max}$; **K 5 (men with BPH), 1 (older women)**
- **PIP bands**: >150 strong; 100–150 normal; 50–100 weak; <50 very weak
- **Detrusor contraction coefficient (DECO)** = $PIP/100$; <1 weak
- **Watts factor (WF_{max})**: <7 W/m² reduced (some 10); volume- and BOO-independent; unvalidated
- **Voluntary stop test** reads ~20 cm H₂O lower than mechanical
- **Continuous occlusion**: most reliable, measures sustainability; painful, research only
- **Detrusor contractility parameter (DCP)** (t20–80): invalid with BOO
- **Shortening velocity** falls before WF — possible early marker
- **Criteria disagree**: DUA diagnosed in 5–55% depending on definition
- **No void on UDS?** → ambulatory UDS: >80% then contract

MANAGEMENT

- **Assess**: diary, DRE, urinalysis, flow, ultrasound PVR, sacral neuro exam, **spinal MRI if deficit**, drug history, constipation
- **Credé / Valsalva**: only DUA + incompetent sphincter (else reflux)
- **ISC**: urethral bleeding ~¼; fewer infections than indwelling catheter
- **Parasympathomimetic** works only if stimulus lacking, not muscle lost; needs **intact sensation**
- **α-Blocker + bethanechol/distigmine** → better flow than α-blocker alone
- **Tamsulosin 0.2 mg**: some women with DUA improve
- **Prostanoids**: PGE → urethral relaxation; **PGF₂α** → **urethral contraction**; uterine side effects
- **ASP8302** (M₃ allosteric modulator) negative; **TAC-302** (neurite growth) ↑ BCI
- **Brindley sacral anterior root stimulator**: complete spinal cord injury (SCI); needs **reflex contractions preop**; + rhizotomy (Sauerwein)
- **Intravesical electrotherapy (IVE)**: Aδ afferents; 10–15 sessions trial; only sham RCT negative

- **Urethral botulinum toxin A (BoNT-A)** 50 U/4 mL: trial-only; best with poor sphincter relaxation
- **Poor outlet-surgery outcome:** Pdet <45, age >80, PVR >1500 mL
- **BOO index (BOOI)** = $P_{det}@Q_{max} - 2Q_{max}$
- **Women:** no bladder neck incision/resection (incontinence, stenosis)
- **Continent catheterisable stoma:** young woman, retention refractory to SNM, declines SPC
- **Ileal conduit:** renal deterioration from thick-walled bladder
- **Latissimus dorsi detrusor myoplasty (LDDM):** inferior epigastric pedicle, intercostal nerve; investigational
- **Reduction cystoplasty:** abandoned
- **Tissue-engineered bladder:** storage only

CLASSIC TRAPS

- **Trap:** UAB has a defining symptom — **Actually:** none is pathognomonic

- **Trap:** low flow + high PVR in man = DUA — **Actually:** could be BOO; needs UDS
- **Trap:** $P_{det}@Q_{max}$ = contraction strength — **Actually:** lowest point on BOR; underestimates
- **Trap:** BOO inevitably decompensates — **Actually:** long-term stable in most men
- **Trap:** acute retention = failed detrusor — **Actually:** detrusor usually preserved
- **Trap:** Parkinson disease → DUA — **Actually:** DO; DUA usually drug-induced
- **Trap:** stroke → DUA — **Actually:** only acute; chronic = DO
- **Trap:** bethanechol treats DUA — **Actually:** unproven; cardiac arrest risk; avoided
- **Trap:** failed void on UDS = acontractile — **Actually:** bashful bladder; ambulatory UDS
- **Trap:** TURP helps symptomatic low-PVR DUA — **Actually:** no better than conservative